ECPS: Cross Pseudo Supervision Based on Ensemble Learning for Semi-Supervised Remote Sensing Change Detection

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***Abstract*— Semi-supervised learning (SSL) aims to exploit the potential of unlabeled data to enhance model performance, which makes it suitable for addressing the challenge of limited labeled data. As a popular technology, pseudo-label is widely applied in many semi-supervised remote sensing (RS) change detection methods. However, when facing limited labeled data, abundant low-quality pseudo-labels from a poorly performing model hinder the effective enhancement of model performance. To address this issue, we propose a novel semi-supervised strategy, named ensemble cross pseudo supervision (ECPS). The utilization of ensemble learning to merge outputs from several change detection models enhances pseudo-label quality, leading to more accurate change information and a significant boost in model performance, even with limited labeled data. In this method, adopting crosswise supervision ensures that no additional inference costs caused by ensemble learning are consumed. This provides both high efficiency and effectiveness for identifying land-cover changes. On the other hand, a simple yet effective ensemble strategy is proposed, which allows to manually adjust the model's tendency toward higher precision or recall for satisfying practical requirements. We conduct extensive experiments on four public RS change detection datasets, and the promising results demonstrate the superiority of the proposed method across various numbers of labeled samples. Our source codes are available at https://github.com/TangXu-Group/ECPS.**

***Index Terms*— Change detection, ensemble learning, pseudo-label, remote sensing (RS), semi-supervised learning (SSL).**

I. Introduction

CHANGE detection plays an important role in the field of remote sensing (RS) image processing. It has been broadly applied into various applications, such as environment monitor, agricultural assessment, and urban expansion [1], [2], [3], [4]. The primary goal of change detection is to identify the land-cover changes between a bitemporal image pair, where two images are captured from the same region but at two different time periods. Recently, advancements in space technology facilitate researchers in acquiring abundant RS images with high spatial resolution (HSR), which offers more details information about land covers. This enables researchers to propose many novel change detection methods with higher accuracy for RS images, greatly accelerating the development of this filed. In particular, by annotating these HSR images to provide much supervision information (i.e., labels), the methods based on deep learning achieve significant performance improvement [5], [6], [7]. However, pursuing high-performing models but regardless of annotation costs proves difficult to meet practical requirements. Therefore, reaching a satisfactory balance between them is a main challenge for current RS change detection task.

At first, traditional RS change detection methods often employ technologies for extracting handcrafted features to identify land-cover changes [8], [9], [10]. These include scale-invariant feature transform (SIFT) [11], slow feature analysis (SFA) [12], and principal component analysis (PCA) [13]. While expertly designed features contribute to efficient implementation, they face challenges with low accuracy and limited generalization. Recently, with the explosive development of deep learning, deep features have been receiving the more and more attention [14], [15], [16]. As two main backbones, convolutional neural networks (CNNs) [17] and Transformers [18] are widely employed to extract deep features from RS images. Taking advantage of their powerful representation for RS images, a lot of methods based on deep features have been proposed, achieving remarkable performance [19], [20], [21]. Nevertheless, as previously mentioned, most of them require abundant labeled data, resulting in huge annotation costs, constraining their applicability.

Consequently, to strike a balance between model performance and annotation cost, researchers introduce semi-supervised learning (SSL) to RS change detection task [22], [23]. While utilizing labeled data to train model in a supervised manner, SSL aims to leverage unlabeled data for further enhancing model performance in an unsupervised manner [24]. Therefore, it is also often used to address the challenge of limited labeled data [25], [26]. To fully exploit the potential of unlabeled data, various advanced technologies or a combination of them are employed for training change detection models, such as adversarial learning [27],

Manuscript received 20 January 2024; accepted 20 February 2024. Date of publication 26 February 2024; date of current version 6 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62171332 and Grant 62276197, in part by the Key Research and Development Program of Shaanxi under Grant 2024GX-YBXM-125, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2024JC-YBMS-472, and in part by Shaanxi Province Innovation Capability Support Plan under Grant 2023-CX-TD-09. *(Corresponding author: Xu Tang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3370236

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consistency learning [28], and pseudo-label [29], [30]. As one of the most popular, pseudo-label technology is applied to several many semi-supervised RS change detection methods. Its effectiveness lies in the introduction of numerous unlabeled data for model training [31], prompting the model to comprehensively learn the distribution of data and thereby enhancing generalization performance. Some of these methods utilize a network to automatically generate pseudo-labels for the unlabeled data and then use them to train this network [32], [33]. Due to the characteristics of self-production and self-use, this training approach is often referred to as self-training [34]. If the quality of the generated pseudo-labels is reasonably satisfactory, self-training can effectively improve the performance of change detection models. However, it encounters difficulties in correcting its own mistakes. Thus, cross training is proposed to solve this problem [35], where pseudo-labels generated by one network are employed to supervise another network. Although cross training has achieved encouraging successes in conducting semi-supervised RS change detection, the crucial precondition for its effectiveness remains acquiring relatively high-quality pseudo-labels to provide precise change information. Here, their quality greatly relies on the initial performance (solely supervised by labeled data) of the model. Hence, when the model is trained on limited labeled data and suffers from a poor initial performance, the low-quality pseudo-labels will result in minimal even negative improvement in model performance.

Therefore, when facing limited labeled data, improving the quality of pseudo-labels becomes a key solution to alleviate this problem. In this article, we propose a semi-supervised strategy for RS change detection task, named ensemble cross pseudo supervision (ECPS), which improves model performance by utilizing unlabeled data in cross training. Ensemble learning, in ECPS, is used for combining several pseudo-labels from different networks to generate a better pseudo-label. Then, the model performance can be significantly improved by using this better pseudo-label, even with limited labeled data. Here, using crosswise supervision makes that we can employ solely a high-preforming model to identify land-cover changes, ensuring no additional costs resulting from ensemble learning are consumed during the inference stage. To conduct semi-supervised RS change detection task, we create a model by embedding ECPS to a simply designed backbone. For labeled data, their labels are simply used to supervise the model training, where we employ cross-entropy function to calculate the loss. Meanwhile, for unlabeled data, ECPS is applied to perform a cross training between different change detection models, thus further enhancing model performance. Moreover, in practical applications, distinct requirements emerge for the model's tendency toward precision or recall due to the typically inverse relationship between them. Therefore, we present a simple yet effective ensemble strategy, which allows for manually adjusting this model's tendency for meeting various practical requirements.

The three main contributions can be summarized as follows.

1) We propose a semi-supervised strategy named ECPS, which employs ensemble learning to improve the pseudo-label quality of unlabeled data. In this way, the model performance can be significantly enhanced, even with limited labeled data.
2) We create a model for conducting the semi-supervised RS change detection task by embedding ECPS in a simply designed backbone. It not only achieves promising performance across various numbers of labeled samples, but also enables manual adjustments for the model's tendency toward precision or recall.
3) We conduct extensive experiments on four public datasets to demonstrate the effectiveness of ECPS and the corresponding RS change detection method.

The rest of this article is organized as follows: 1) Section II reviews the existing semi-supervised works; 2) the preliminary and the proposed method are introduced in Section III; 3) experimental results are presented in Section IV; and 4) in Section V, we summarize this article and discuss future works.

II. Related Work

In this section, we provide a comprehensive review of existing works related to the semi-supervised task. These works are categorized into two main parts: 1) semi-supervised methods for RS change detection and 2) other notable semi-supervised methods.

A. Semi-Supervised Methods for RS Change Detection

Fully exploiting the potential of unlabeled data to enhance model performance lies at the heart of conducting the semi-supervised RS change detection task. Fortunately, adversarial learning has inherent advantages in handling such unlabeled data [23], [27], [36]. For example, Gong et al. [27] proposed a generative discriminatory classified network (GDCN), where labeled, unlabeled, and fake data (obtained from generator) were simultaneously employed to train the discriminator for generating the predicted change map, thereby boosting its capacity in detecting changes. In addition, a semi-supervised convolutional network [23], referred to as SemiCDNet, was proposed based on the generative adversarial network (GAN). SemiCDNet utilized two discriminators to enhance the distribution consistency of segmentation and entropy maps between labeled and unlabeled data. Besides the adversarial learning, the consistency learning also plays an important role in utilizing unlabeled data to enhance the model performance. In the literature by Bandara and Patel [37], for the unlabeled data, they applied small random perturbations to the original feature maps to construct two perturbed feature maps. By enhancing the consistency between the original feature map and two perturbed feature maps, the model performance was further improved. In addition to utilizing feature consistency, the multitask consistency (MTC) was proposed to construct an MTC network (MTCNet) [28], which simultaneously conducted the building segmentation and semi-supervised RS change detection tasks. Then, by improving the result consistency of these two tasks, the generalization ability of model is boosted.

The presence of numerous unlabeled data is often due to the high cost of manual annotation. Therefore, another common approach to leverage unlabeled data is by automatically

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generating pseudo-labels for them [29], [34], [38]. For example, a semi-supervised adaptive ladder network (SSALN) [34] was proposed to generate pseudo-labels for exploiting implicit information, where the continuous iterations were employed to make the pseudo-labels more reliable and accurate. Similarly, Wang et al. [38] proposed a reliable contrastive learning (RCL) method, which utilized the uncertainty of unlabeled data to obtain reliable pseudo-labels for training the model. Furthermore, the pseudo-labels were also utilized to enhance the model understanding of land covers within bitemporal images. To achieve this, a several semi-supervised change detection (SSCD) method was developed [29]. SSCD leveraged the pseudo-labels generated from weakly augmented images to supervise the prediction of strong-augmented images.

B. Other Notable Semi-Supervised Methods

There are also several notable semi-supervised methods for other fields, which have shown promising performance and versatility [35], [39], [40]. Similar to the semi-supervised RS change detection task, technologies used by these methods can be broadly divided into three groups: adversarial learning, consistency learning, and pseudo-label. For example, a semi-supervised semantic segmentation GAN (s4GAN) was proposed [39], in which adversarial learning with feature matching loss was used to exploit the unlabeled data. In addition, a fully convolutional discriminator [41] was designed to identify the reliable regions from the predictions of unlabeled data for performing SSL. In general, the highly accurate prediction map corresponds to a low entropy map [40]. Thus, researchers proposed a method with adversarial entropy minimization, referred to as ADVENT [40]. For methods based on consistency learning, data augmentation often is employed to generate different representations of the same image for conducting consistency training. In the literature [42], FixMatch is proposed for unlabeled data, which applies the results of weakly augmentation to supervise the predictions of strongly augmentation. In this way, the consistency between these two augmentations is enhanced, resulting in an improved model performance. Likewise, Xie et al. [55] proposed using more robust data augmentation strategy to generate various images. Subsequently, the consistency training is performed between original image and its augmented versions. In addition to using data augmentation to obtain images with different versions, the mean teacher (MT) approach can also be employed to similar effectiveness (i.e., different feature versions of the same image) [43]. Enhancing the consistency between these different feature versions can further exploit valuable information within unlabeled data.

Pseudo-labels have proven to be highly effective in assisting models in utilizing unlabeled data [35], [44], [45]. In [44], researchers introduced a Tri-Net model, where a pseudo-label editing strategy is used to obtain stable pseudo-labels. Then, they were incorporated into the training data as the ground truth of the unlabeled data to train the model. Moreover, a semi-supervised RS image semantic segmentation method, referred to as RanPast, was proposed [45]. RanPast utilized the predictions of unlabeled data and the ground truth of labeled data to generate pseudo-labels at the patch level. Pseudo-label technology is also employed in crosswise supervision. To achieve accurate results, the pseudo-labels of unlabeled data, generated by two networks with the same structure but different initial parameters, were utilized to conduct cross pseudo supervision (CPS) [35]. In fact, the mentioned methods generally do not rely solely on a single semi-supervised technology. Instead, they usually combine different technologies, such as pseudo-label and consistency learning, to achieve outstanding performance.

III. Proposed Approach

In semi-supervised task, the given dataset $\mathcal{D}$ comprises labeled data $\mathcal{D}^l$ and the unlabeled data $\mathcal{D}^u$. The labeled data $\mathcal{D}^l$ can be easily used to train the model in a supervised manner, with each network in our method being supervised using the cross-entropy function. Therefore, in the following content, we only focus on how to use the unlabeled data $\mathcal{D}^u$ (simplified as $\mathbf{X}$) to further improve the model performance in an unsupervised manner. Note that in our experiments, the training of labeled and unlabeled data is conducted simultaneously.

First, we review four popular self-supervised strategies in Section III-A, including CPS [35], cross confidence consistency (CCC) [46], MT [47], and PseudoSeg (PS) [48]. Based on the given respective descriptions, they are reproduced and applied to our backbone model, which allows us to compare and evaluate the effectiveness of the proposed ECPS in the experimental part. Subsequently, we will introduce the details of ECPS and the prediction ensemble strategy in Sections III-B and III-C, respectively.

A. Preliminary

To clearly demonstrate four strategies, we simplify the data processing of model as follows:

$$\mathbf{X} \rightarrow f(\theta) \rightarrow \mathbf{P} \nrightarrow \mathbf{Y} \tag{1}$$

where $\mathbf{X}$ denotes the inputted image, θ represents the model's parameter, $\mathbf{P}$ represents the outputted prediction result after the softmax normalization layer, $\mathbf{Y}$ corresponds to the one-hot pseudo-label generated by discretizing the values of $\mathbf{P}$, and $\nrightarrow$ means "no backpropagation."

1) Cross Pseudo Supervision: CPS adopts two models to conduct crosswise supervision, both sharing the same structure but initialized with different parameters, i.e.,

$$\begin{matrix} & \nearrow f(\theta_1) \rightarrow \mathbf{P}_1 \nrightarrow \mathbf{Y}_1 \\ \mathbf{X} & \\ & \searrow f(\theta_2) \rightarrow \mathbf{P}_2 \nrightarrow \mathbf{Y}_2. \end{matrix} \tag{2}$$

Next, based on a crosswise mode, the pseudo-labels $\mathbf{Y}_1/\mathbf{Y}_2$ are used to supervise the predictions $\mathbf{P}_2/\mathbf{P}_1$, respectively. This process is visually shown in Fig. 1(a). The corresponding loss $\mathcal{L}_{\text{cps}}$ can be calculated as follows:

$$\mathcal{L}_{\text{cps}} = \frac{1}{W \times H} \sum_{i=1}^{W \times H} \left(f_{\text{ce}}(\mathbf{p}_{2i}\mathbf{y}_{1i}) + f_{\text{ce}}(\mathbf{p}_{1i}, \mathbf{y}_{2i}) \right) \tag{3}$$

where W and H denote the width and height of image $\mathbf{X}$, $\mathbf{p}_{*i}$ and $\mathbf{y}_{*i}$ severally represent the ith pixel value of $\mathbf{P}_*$ and $\mathbf{Y}_*$, and $f_{\text{ce}}(\cdot)$ is the cross-entropy function.

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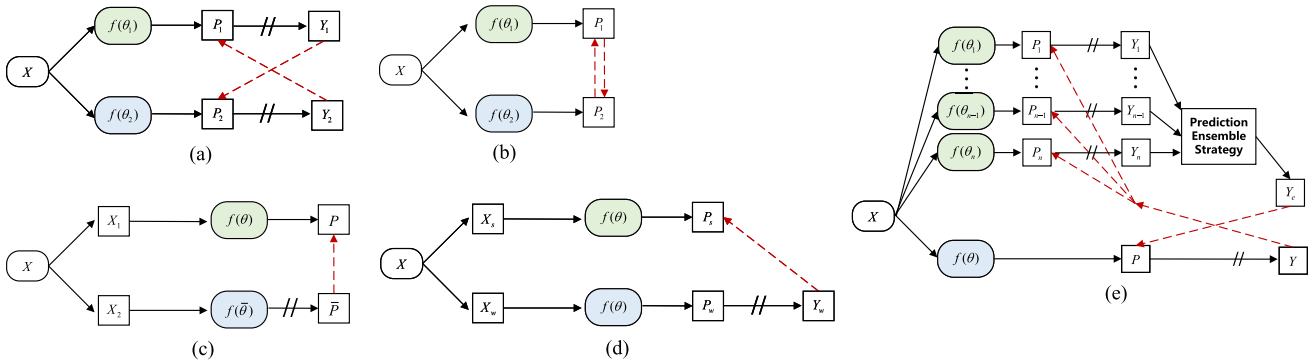


Fig. 1. Framework of five semi-supervised strategies. (a) CPS [35]. (b) CCC [46]. (c) MT [47]. (d) PS [48]. (e) Our ECPS. Here, in ECPS, the green/blue marked model is referred to as student/teacher for clear distinction.

2) Cross Confidence Consistency: Similar to CPS, CCC also adopts a crosswise supervision mode, where two models have the same structure but different initial parameters. However, instead of utilizing pseudo-labels $\mathbf{Y}$, it directly treats the $\mathbf{P}$ as the supervision target [see Fig. 1(b)], i.e.,

$$\mathbf{X} \begin{cases} \nearrow f(\theta_1) \to \mathbf{P}_1 \\ \searrow f(\theta_2) \to \mathbf{P}_2. \end{cases} \tag{4}$$

Here, the loss $\mathcal{L}_{\text{ccc}}$ is calculated according to the following equation:

$$\mathcal{L}_{\text{ccc}} = \frac{1}{W \times H} \sum_{i=1}^{W \times H} \left(f_{l2}(\mathbf{p}_{2i}, \mathbf{p}_{1i}) + f_{l2}(\mathbf{p}_{1i}, \mathbf{p}_{2i}) \right) \tag{5}$$

where $f_{l2}(\cdot)$ is the function for computing the $L2$ norm, and the meanings of other characters are the same as those in (3).

3) Mean Teacher: In MT, two models with the same structure are constructed, including a teacher model and a student model. Then, the prediction of teacher model is used to supervise the student model, as shown in Fig. 1(c). Here, the parameters of student model are initialized randomly. Specifically, the initial parameters of teacher model are set to the same as those of student model and then updated based on the exponential moving average (EMA) [49]. Its logical form is

$$\begin{aligned} &\mathbf{X} \begin{cases} \nearrow \mathbf{X}_1 \to f(\theta) \to \mathbf{P} \\ \searrow \mathbf{X}_2 \to f(\overline{\theta}) \nrightarrow \overline{\mathbf{P}} \end{cases} \\ &\overline{\theta}_t = \lambda \theta_t + (1 - \lambda) \overline{\theta}_{t-1}, t \in [0, n_i) \end{aligned} \tag{6}$$

where $\mathbf{X}_1$ and $\mathbf{X}_2$ are two distinct augmented versions of $\mathbf{X}$, θ and $\overline{\theta}$ separately denote the parameters of student and teacher models, θ_t and $\overline{\theta}_t$ represent the parameters of student and teacher models in the tth training iteration, n_i equals the number of training iterations, and λ is the hyperparameter for regulating the impact of previous parameters, which is set to be 0.999 [47]. It is worth noting that "no backpropagation" ($\nrightarrow$) is adopted in the teacher branch, since the parameters of teacher model are not updated through backpropagation. The loss $\mathcal{L}_{\text{mt}}$ can be calculated as follows:

$$\mathcal{L}_{\text{mt}} = \frac{1}{W \times H} \sum_{i=1}^{W \times H} f_{l2}(\mathbf{p}_i, \overline{\mathbf{p}_i}) \tag{7}$$

where $\mathbf{p}_i$ and $\overline{\mathbf{p}_i}$ are from the student and teacher branches, respectively, and the others refer to (3) and (5).

4) PseudoSeg: All three previous strategies conduct an unsupervised manner based on two models with different parameters. However, PS achieves a successful attempt in optimizing data augmentation based on two models with sharing parameters. Unlike the simple data augmentation used in MT, where two slightly different versions, $\mathbf{X}_1$ and $\mathbf{X}_2$, are created from $\mathbf{X}$, PS adopts a more ingenious strategy. As illustrated in Fig. 1(d), it applies both weak and strong augmentations to $\mathbf{X}$ for generating $\mathbf{X}_w$ and $\mathbf{X}_s$, respectively. Then, the prediction results of $\mathbf{X}_w$ are used to generate the pseudo-label $\mathbf{Y}_w$ for supervising the prediction $\mathbf{P}_s$ of $\mathbf{X}_s$, i.e.,

$$\mathbf{X} \begin{cases} \nearrow \mathbf{X}_s \to f(\theta) \to \mathbf{P}_s \\ \searrow \mathbf{X}_w \to f(\theta) \to \mathbf{P}_w \nrightarrow \mathbf{Y}_w. \end{cases} \tag{8}$$

Therefore, the calculation of the loss $\mathcal{L}_{ps}$ is as follows:

$$\mathcal{L}_{ps} = \frac{1}{W \times H} \sum_{i=1}^{W \times H} f_{\text{ce}}(\mathbf{p}_{si}, \mathbf{y}_{wi}) \tag{9}$$

where $\mathbf{p}_{si}$ and $\mathbf{y}_{wi}$ denote the ith pixel value of $\mathbf{P}_s$ and $\mathbf{Y}_w$, respectively.

B. Ensemble Cross Pseudo Supervision

Let us further analyze why CPS is effective and what potential problem it arises. As mentioned above, CPS adopts the crosswise supervision of pseudo-label mechanism between two models. Thus, we take the model $f(\theta_1)$ as an example for explanation, and the same situation will happen in model $f(\theta_2)$. CPS makes that model $f(\theta_1)$ can learn much knowledge (contains right or wrong) from the pseudo-label $\mathbf{Y}_2$ generated by the other model $f(\theta_2)$. Here, wrong knowledge corresponds to the inevitable noises within $\mathbf{Y}_2$. Therefore, when the noises are slight (more right knowledge), the performance of model $f(\theta_1)$ can be improved significantly. In this scenario, CPS is effective, which generally corresponds to that the moderate/many labeled data $\mathcal{D}^l$ lead to a balanced/promising performance of model $f(\theta_2)$. However, when limited labeled data $\mathcal{D}^l$ cause a poor performance of model $f(\theta_2)$, numerous noises within $\mathbf{Y}_2$ (more wrong knowledge) may even collapse the performance of model $f(\theta_1)$, which has been proved in

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our experiments. Therefore, the key to alleviate the problem of CPS is reducing noises within pseudo-labels, i.e., improving their quality. In this way, the model performance will be more stable and powerful, even with limited labeled data $\mathcal{D}^l$.

To achieve this, ensemble learning is employed in the proposed ECPS, which can combine the pseudo-labels of several models to generate a better pseudo-label for improving model performance, even with limited labeled data. In addition, when facing moderate or abundant labeled data, the ECPS also can make the model obtains stronger performance than CPS, by generating more reliable pseudo-labels. The structure of ECPS is depicted in Fig. 1(e). In ECPS, the model $f(\theta_i)$ involves ensemble, which is called the student model, while the individual model $f(\theta)$ (to be used during the inference stage) is referred to as the teacher model for clarity. The corresponding data processing as follows:

$$\begin{array}{ccccccc} & \nearrow & f(\theta_1) \rightarrow \mathbf{P}_1 \rightarrow \mathbf{Y}_1 & \searrow & \\ \mathbf{X} \rightarrow & \cdots & \cdots & \cdots & \rightarrow \mathbf{Y}_e \\ \nearrow & \searrow & f(\theta_n) \rightarrow \mathbf{P}_n \rightarrow \mathbf{Y}_n & \nearrow & \\ \mathbf{X} & & & & \\ \searrow & \mathbf{X} \rightarrow & f(\theta) \rightarrow \mathbf{P} \rightarrow \mathbf{Y} & & \end{array} \tag{10}$$

where θ_i is the parameter of the ith ensemble student model, θ denotes the parameter of the teacher model, $\mathbf{Y}_e$ corresponds to the ensemble pseudo-label of student models generated by ensemble function (further explained in Section III-C), and $\mathbf{Y}$ represents the pseudo-label of the teacher model. Then, the crosswise supervision between $\mathbf{Y}_e$ and $\mathbf{Y}$ is conducted, and the loss $\mathcal{L}_{\text{ecps}}$ can be calculated as follows:

$$\mathcal{L}_{\text{ecps}} = \frac{1}{W \times H} \sum_{i=1}^{W \times H} \left(f_{\text{ce}}(\mathbf{p}_i, \mathbf{y}_{ei}) + \sum_{j=1}^{n} f_{\text{ce}}(\mathbf{p}_{ji}, \mathbf{y}_i) \right) \tag{11}$$

where n equals the number of student models, $\mathbf{p}_i$ and $\mathbf{p}_{ji}$ are the ith pixel value of $\mathbf{P}$ and $\mathbf{P}_j$, and $\mathbf{y}_{ei}$ and $\mathbf{y}_i$ denote the ith pixel value of $\mathbf{Y}_e$ and $\mathbf{Y}$.

Under the supervision of ensemble pseudo-label $\mathbf{Y}_e$, the teacher model $f(\theta)$ is capable of achieving the stronger performance than the single student model $f(\theta_i)$. As a result, during the inference stage, the final predictions are based on the outputs of the teacher model, while student models will be removed. This skill allows us to eliminate the computational costs associated with ensemble learning while generating the high-accuracy results through the teacher model.

C. Prediction Ensemble Strategy

To combine the pseudo-labels of all student models from $\mathbf{Y}_1$ to $\mathbf{Y}_n$, first, they are averaged to generate the transition pseudo-label $\mathbf{Y}'_e$ at pixel level, i.e., $\mathbf{y}'_{ei} = (1/n)\sum_{j=1}^{n} \mathbf{y}_{ji}$, where $\mathbf{y}'_{ei}$ and $\mathbf{y}_{ji}$ represent the ith pixel value of $\mathbf{Y}'_e$ and $\mathbf{Y}_j$, and n denotes the number of student models. Then, an ensemble function f_{ensemble} is applied to readjust the value of $\mathbf{Y}'_e$ as the final pseudo-label $\mathbf{Y}_e$, providing supervision

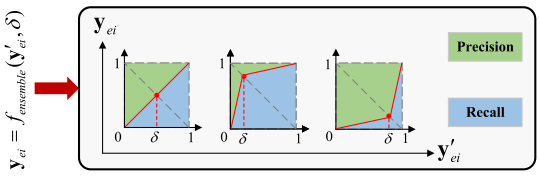


Fig. 2. Function f_{ensemble} for balancing precision and recall. Here, the size of the green or blue areas corresponds to the model's tendency to achieve higher precision or recall.

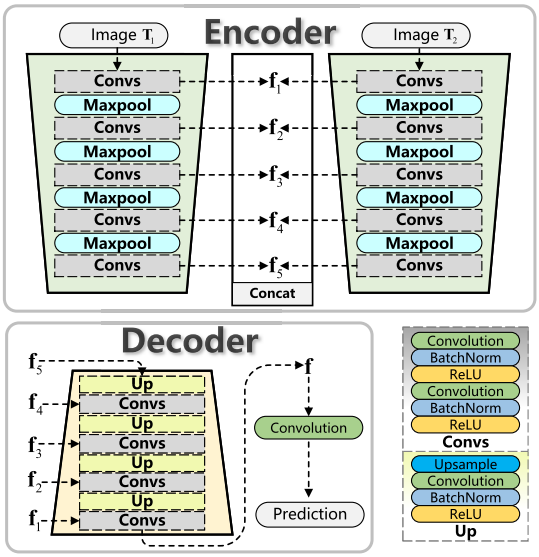


Fig. 3. General RS change detection model for receiving bitemporal image pair $\mathbf{T}_1$ and $\mathbf{T}_2$ and then generating predicted change results.

information for teacher model, given by

$$\mathbf{y}_{ei} = f_{\text{ensemble}}(\mathbf{y}'_{ei}, \delta) = \begin{cases} \dfrac{1-\delta}{\delta} \times \mathbf{y}'_{ei}, & \mathbf{y}'_{ei} \le \delta \\ \dfrac{\delta}{1-\delta} \times \mathbf{y}'_{ei} + \dfrac{1-2\cdot\delta}{1-\delta}, & \mathbf{y}'_{ei} > \delta \end{cases} \tag{12}$$

where $\mathbf{y}_{ei}$ is the ith pixel value of $\mathbf{Y}_e$ and δ is the hyperparameter with a value range of (0, 1) for balancing precision and recall. Note that δ must not equal 0 or 1 to avoid generating an invalid pseudo-label $\mathbf{Y}_e$, which consists entirely of zeros or ones. The visual representation of f_{ensemble} is shown in Fig. 2, where three samples are given with δ values less than, equal to, and more than 0.5. According to the figure, f_{ensemble} is composed of two lines segmented at the point $[\delta, 1-\delta]$. Consequently, smaller/larger values of δ will adjust the transition pseudo-label $\mathbf{y}'_{ei}$ closer to 1/0 (change/unchange), resulting in higher recall/precision, respectively.

D. Backbone

A general RS change detection model is designed as our backbone, which receives bitemporal images $\mathbf{T}_1$ and $\mathbf{T}_2$ and generates the predicted change results. The visual model framework is shown in Fig. 3, which adopts a frequently used encoder–decoder structure. In encoder, five convolution groups ("Convs") and four maxpool layers are utilized, where each "Convs" is composed of convolution with the kernel size of 3 × 3, BatchNorm, and rectified linear unit (ReLU) layers. Similarly, encoder consists of "Convs" and "Up," where

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"Up" adds the upsample layer with bilinear interpolation. The maxpool and upsample layers have a scale factor of 2 for reducing and increasing feature size, respectively. After obtaining bitemporal images $\mathbf{T}_1$ and $\mathbf{T}_2$, they are first inputted into parameter-shared encoder, and then, their features from five "Convs" are concatenated to five features from $\mathbf{f}_1$ to $\mathbf{f}_5$. Second, these five features are inputted into the decoder, in turn, from $\mathbf{f}_5$ to $\mathbf{f}_1$ to generate the final feature $\mathbf{f}$. Finally, a convolution layer with the kernel size of 1×1 is employed to generate prediction results $\mathbf{P}$.

IV. Experiments

A. Experimental Settings

1) Datasets: We select four public datasets to evaluate the model performance, including learning vision and remote sensing (LEVIR) [3], season-varying change detection (SVCD) [50], Sun Yat-sen University (SYSU) [51], and Onera Satellite Change Detection (OSCD) [52]. LEVIR [3] is challenging RS change detection dataset, where images are collected from Google Earth platform. It consists of 637 image pairs, each with the image size of 1024×1024, the spatial resolution of 0.5 m per pixel, spanning from 2002 to 2018. For convenience, we crop these images from $1 \times 1024 \times 1024$ to $16 \times 256 \times 256$, which is a frequently used setting for LEVIR dataset [37]. Finally, according to the data partitioning of original paper, these cropped images with the size of 256×256 are divided into training (7120 pairs), validation (1024 pairs), and testing (2048 pairs) sets. SVCD [50] contains 16k image pairs, each with the size of 256×256 and the spatial resolution ranging from 3 to 100 cm per pixel. The dataset is divided into 10k pairs for training, 3k pairs for validation, and 3k pairs for testing. These images are cropped from seven real image pairs with the size of 4725×2700 and four synthetic image pairs (where additional objects were manually added) with the size of 1900×1000. SYSU [51] is a dataset of RS images acquired from Hong Kong, spanning the years from 2007 to 2014. It consists of 20k bitemporal image pairs, each measuring 256×256 pixels with a spatial resolution of 0.5 m per pixel. According to the ratio of 6:2:2, the dataset is split into training, validation, and testing sets with 12k, 4k, and 4k image pairs, respectively. OSCD [52] comes from Sentinel-2 satellite observations, covering 24 cities at two different time points. The training/test set, as per the original paper [52], consists of image pairs from 14/10 cities, each containing 13 spectral channels. According to the literature [53], we crop images to 32×32 with a stride of 6 and retain images that include both changed and unchanged pixels.

2) Metrics: In order to comprehensively evaluate our method, we adopt five metrics in this article: precision (Pre), recall (Rec), $F1$ score ($F1$), intersection over union (IoU), and overall accuracy (OA). Their definitions are as follows:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}} \\ F1 &= \frac{2 \cdot \text{Pre} \cdot \text{Rec}}{\text{Pre}+\text{Rec}}, \quad \text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \end{aligned} \tag{13}$$

where TP, FP, TN, and FN refer to the pixel counts of true positive, false positive, true negative, and false negative, respectively. Among five metrics, $F1$, IoU, and OA provide a holistic view of the model's performance. On the other hand, Pre and Rec are two biased metrics that are often inversely proportional to each other. Specifically, Pre measures the accuracy of pixels predicted as change, while Rec indicates the accuracy of pixels correctly identified as change among all the actual change pixels.

3) Implementation Details: Based on deep framework PyTorch [54], all experiments are conducted on a server equipped with two RTX 3090 GPUs and an Inter Xenon Silver 4214R CPU. During training, we adopt the Adam optimizer with the learning rate of 0.0001 to optimize model parameters and use the OneCycleLR scheduler with the annealing strategy of "cos" to update the learning rate. The batch size and the epoch number are set to 16 and 99, respectively. The model parameters are initialized randomly. In addition, we employ three simple strategies for extending the dataset, including random cropping, rotation, and flipping. According to the experimental results (see Section IV-C), two hyperparameters, δ and the student number n, are set to 0.3 and 3 for obtaining the model with the best performance, respectively.

B. Comparison Studies

To demonstrate the superiority of our ECPS, we compare it with four advanced semi-supervised methods: RCL [38], s4GAN [39], AdvNet [41], and Revisiting [37]. In addition, it is worth noting that our method consists of a backbone and the proposed semi-supervised strategy ECPS. Thus, conducting a method comparison solely may not fully demonstrate the superiority of ECPS, as the performance can be influenced by the capabilities of our backbone model. Therefore, to ensure a fair evaluation, we also adopt four popular semi-supervised strategies (see Section III-A): CPS, CCC, MT, and PS. These strategies are incorporated into our backbone to conduct the semi-supervised RS change detection task under the same experimental settings as ECPS.

All experiments about the method and strategy comparisons are performed on four datasets (LEVIR, SVCD, SYSU, and OSCD) with four different training ratios (5%, 10%, 20%, and 40%). In each case, $x\%$ of training set is as the labeled data $\mathcal{D}_l$ for training model in a supervised manner, while the remaining $(1-x\%)$ of the training set is as the unlabeled data $\mathcal{D}_u$ for the unsupervised training without labels. Note that, we adopt $F1$, IoU, and OA to evaluate the overall performance, while Pre and Rec are given to explore model's tendency.

1) Method Comparison: The method comparison results are shown in Table I. It is evident that our method achieves the best performance on three datasets, LEVIR, SVCD, and OSCD, across four different training ratios. Notably, at a training ratio of 5%, our method achieves an outstanding $F1$/IoU/OA, surpassing the lowest results by 6.18%/9.03%/0.56% on LEVIR, 7.09%/9.62%/1.3% on SVCD, and 4.51%/4.11%/5.9% on OSCD, representing a significant improvement. These results demonstrate the stability and strength of our method, even when dealing with limited labeled data. For the SYSU dataset, our method obtains superior accuracy at training ratios of 10%

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TABLE I

COMPARISON STUDIES OF DIFFERENT SEMI-SUPERVISED METHODS ON DIFFERENT TRAINING RATIOS OF FOUR DATASETS

Ratio	Methods	LEVIR					SVCD					SYSU					OSCD				
		Pre	Rec	**F1**	**IoU**	**OA**	Pre	Rec	**F1**	**IoU**	**OA**	Pre	Rec	**F1**	**IoU**	**OA**	Pre	Rec	**F1**	**IoU**	**OA**
40%	RCL [38]	89.24	82.32	85.64	74.89	98.59	87.37	88.15	87.76	78.19	96.82	70.88	82.86	76.40	61.81	88.93	70.42	47.86	56.99	39.85	84.44
	s4G. [39]	84.41	88.45	86.38	76.02	98.58	92.95	71.69	79.64	65.23	95.41	95.41	70.02	76.43	61.84	89.81	72.50	45.21	55.70	38.59	84.45
	Adv. [41]	90.63	85.14	87.80	78.25	98.79	93.78	88.79	91.22	83.85	97.88	80.66	77.20	**78.89**	**65.14**	**90.26**	69.28	46.35	55.54	38.45	86.54
	Rev. [37]	90.83	85.84	88.26	78.99	98.84	88.30	83.71	85.94	75.35	96.46	80.34	76.18	78.21	64.21	89.99	69.05	47.83	56.51	39.39	84.15
	Our	90.46	86.54	**88.46**	**79.30**	**98.85**	94.78	89.46	**92.04**	**85.25**	**98.08**	80.84	74.08	77.31	63.02	89.75	61.04	56.34	**58.60**	**41.44**	**88.33**
20%	RCL [38]	86.61	82.54	84.65	73.38	98.47	83.26	83.52	83.39	71.52	95.70	69.27	84.35	76.07	61.38	88.48	56.24	54.69	55.45	38.36	81.08
	s4G. [39]	85.42	84.67	85.04	73.98	98.48	89.68	69.98	78.62	64.77	95.08	75.63	75.24	75.43	60.56	88.44	72.40	42.58	53.62	36.63	82.71
	Adv. [41]	91.79	82.55	86.92	76.87	98.73	93.50	88.10	90.71	83.01	97.76	81.47	73.10	77.06	62.68	**89.73**	64.28	49.56	55.97	38.86	86.23
	Rev. [37]	91.75	83.71	87.55	77.85	**98.79**	88.54	83.08	85.73	75.02	96.42	76.96	67.40	71.86	56.08	87.55	55.38	55.23	55.30	38.22	80.78
	Our	90.85	84.47	**87.68**	**78.06**	98.70	94.65	89.25	**91.87**	**84.97**	**98.04**	79.41	75.77	**77.55**	**63.33**	89.65	59.26	55.76	**57.46**	**40.31**	**87.89**
10%	RCL [38]	87.73	81.34	84.42	73.03	98.47	81.29	82.31	81.80	69.20	95.27	82.82	69.68	75.10	60.42	87.48	57.91	51.79	54.68	37.62	81.52
	s4G. [39]	86.20	80.58	83.25	71.30	98.35	86.12	71.83	78.33	64.38	94.86	77.50	68.35	72.64	57.03	87.56	68.36	40.92	51.20	34.41	81.96
	Adv. [41]	86.03	83.24	84.61	73.33	98.46	89.39	82.30	85.70	74.97	96.59	77.44	73.31	75.32	60.41	88.67	58.64	49.95	53.94	36.93	85.32
	Rev. [37]	91.42	83.13	87.08	77.12	98.74	86.34	80.09	83.10	71.08	95.79	79.61	59.56	68.14	51.68	86.87	51.77	57.34	54.41	37.38	79.31
	Our	89.30	85.59	**87.40**	**77.63**	**98.74**	92.57	84.36	**88.27**	**79.01**	**97.22**	77.25	75.96	**76.60**	**62.07**	**89.05**	52.05	57.74	**54.75**	**37.69**	**86.01**
5%	RCL [38]	88.99	75.52	81.70	69.07	98.28	84.08	75.95	79.81	66.40	95.03	68.70	75.97	72.15	56.44	86.17	78.52	38.97	52.09	35.21	84.56
	s4G. [39]	83.48	76.61	79.90	66.53	98.03	91.43	63.55	74.98	59.98	94.52	78.22	61.81	69.05	52.73	86.94	71.62	38.68	50.24	33.54	80.85
	Adv. [41]	85.52	76.70	80.87	67.89	98.15	85.79	73.01	78.88	65.13	95.14	75.72	72.28	**73.96**	**58.68**	**88.00**	56.95	43.86	49.56	32.94	86.02
	Rev. [37]	91.37	79.79	85.18	74.19	89.59	80.95	75.11	77.92	63.83	94.50	70.50	64.20	67.20	50.61	85.22	58.67	47.29	52.37	35.47	81.48
	Ours	86.61	85.55	**86.08**	**75.56**	**98.59**	87.75	77.09	**82.07**	**69.60**	**95.82**	70.09	75.78	72.82	57.26	86.66	54.99	53.18	**54.07**	**37.05**	**86.75**

TABLE II

COMPARISON STUDIES OF DIFFERENT SEMI-SUPERVISED STRATEGIES ON DIFFERENT TRAINING RATIOS OF FOUR DATASETS. HERE, "SUPONLY" SERVES AS THE BASELINE, REPRESENTING THE RESULTS OF THE BACKBONE TRAINED SOLELY USING LABELED DATA $\mathcal{D}_l$

Ratio	Strategies	LEVIR					SVCD					SYSU					OSCD				
		Pre	Rec	**F1**	**IoU**	**OA**	Pre	Rec	**F1**	**IoU**	**OA**	Pre	Rec	**F1**	**IoU**	**OA**	Pre	Rec	**F1**	**IoU**	**OA**
40%	SupOnly	88.65	83.55	86.02	75.47	98.62	92.28	86.53	89.31	80.69	97.43	74.22	74.22	74.22	59.01	87.84	56.77	53.53	55.10	38.03	87.21
	+ PS [48]	87.46	86.66	87.06	77.08	98.69	92.13	85.50	88.69	79.68	97.29	75.20	77.67	76.42	61.84	88.69	61.60	50.44	55.47	38.38	88.12
	+ MT [47]	91.54	83.03	87.07	77.11	98.74	93.62	87.41	90.41	82.50	97.70	78.28	71.72	74.85	59.81	88.64	64.26	47.14	54.38	37.35	88.40
	+ CCC [46]	90.07	83.83	86.84	76.74	98.71	93.42	87.05	90.12	82.02	97.63	79.70	72.65	76.01	61.31	89.19	63.93	47.26	54.35	37.31	88.36
	+ CPS [35]	91.22	85.60	88.32	79.09	98.85	94.85	88.84	91.75	84.75	98.01	79.71	73.45	76.45	61.88	89.33	66.57	49.86	57.02	39.88	88.33
	+ Ours	90.46	86.54	**88.46**	**79.30**	**98.85**	94.78	89.46	**92.04**	**85.25**	**98.08**	80.84	74.08	**77.31**	**63.02**	**89.75**	61.04	56.34	**58.60**	**41.44**	**88.98**
20%	SupOnly	90.60	80.87	85.46	74.61	98.60	90.98	87.15	89.03	80.22	97.33	79.56	68.43	73.57	58.19	88.41	55.20	50.70	52.86	35.92	86.74
	+ PS [48]	86.62	85.71	86.16	75.69	98.60	90.52	86.91	88.68	79.66	97.24	78.93	72.26	75.45	60.58	88.91	57.97	50.12	53.76	36.76	87.36
	+ MT [47]	90.97	81.85	86.17	75.71	98.66	91.48	86.49	88.91	80.04	97.32	76.00	70.50	73.15	57.66	**89.79**	59.96	45.08	51.47	34.65	87.53
	+ CCC [46]	90.88	82.05	86.24	75.80	98.67	93.07	86.97	89.92	81.68	97.58	72.23	75.92	74.03	58.77	87.44	60.28	47.63	53.22	36.25	87.72
	+ CPS [35]	92.22	83.14	87.44	77.69	**98.78**	94.79	87.62	91.06	83.59	97.86	84.22	67.32	74.83	59.78	89.32	64.56	49.24	55.87	38.76	87.89
	+ Ours	90.85	84.47	**87.68**	**78.06**	98.70	94.65	89.25	**91.87**	**84.97**	**98.04**	79.41	75.77	**77.55**	**63.33**	89.65	59.26	55.76	**57.46**	**40.31**	**88.59**
10%	SupOnly	87.47	78.15	82.55	70.28	98.32	91.02	79.13	84.66	73.40	96.44	76.45	68.87	72.46	56.82	87.66	59.63	44.88	51.21	34.42	87.46
	+ PS [48]	88.68	83.94	86.25	75.82	98.64	89.42	83.37	86.29	75.89	96.71	77.10	74.11	75.58	60.74	88.70	57.56	44.85	50.42	33.71	87.07
	+ MT [47]	89.12	80.79	84.75	73.54	98.52	89.18	82.00	85.44	74.58	96.53	79.48	65.29	71.69	55.87	87.84	53.87	42.49	47.51	31.15	86.23
	+ CCC [46]	92.94	73.16	81.87	69.31	98.35	89.34	75.22	81.67	69.02	95.81	79.36	59.72	68.15	51.69	86.84	61.33	39.01	47.69	31.31	87.45
	+ CPS [35]	92.13	79.33	85.25	74.30	98.60	93.52	78.44	85.32	74.40	96.65	80.55	63.25	70.86	54.87	87.73	66.79	37.89	48.35	31.88	86.01
	+ Ours	89.30	85.59	**87.40**	**77.63**	**98.74**	92.57	84.36	**88.27**	**79.01**	**97.22**	77.25	75.96	**76.60**	**62.07**	**89.05**	52.05	57.74	**54.75**	**37.69**	**88.13**
5%	SupOnly	86.90	75.04	80.53	67.41	98.15	86.49	70.98	77.97	63.90	95.02	77.15	64.81	70.44	54.37	87.17	62.58	30.30	40.83	25.65	86.75
	+ PS [48]	86.69	82.99	84.80	73.61	98.48	80.47	77.92	79.18	65.53	94.91	73.35	73.85	**73.60**	**58.23**	87.51	54.23	37.44	44.30	28.45	86.19
	+ MT [47]	91.40	73.49	81.47	68.73	98.30	84.62	74.23	79.08	65.40	95.12	78.98	65.93	71.87	56.09	**87.83**	49.69	40.27	44.48	28.60	85.26
	+ CCC [46]	91.44	64.68	75.77	60.99	97.89	91.97	52.35	66.72	50.06	93.51	77.42	59.40	67.22	50.63	86.34	34.97	18.25	23.98	13.63	83.04
	+ CPS [35]	92.23	68.31	78.49	64.60	98.09	93.80	56.80	70.76	54.75	94.17	74.44	57.01	64.57	47.68	85.25	41.25	16.35	23.41	13.26	84.32
	+ Ours	86.61	85.55	**86.08**	**75.56**	**98.59**	87.75	77.09	**82.07**	**69.60**	**95.82**	70.09	75.78	72.82	57.26	86.66	54.99	53.18	**54.07**	**37.05**	**87.12**

and 20%. Although the second-best performance is achieved at the training ratios of 5% and 40%, the slight 1% accuracy difference proves its advancement. In summary, the proposed method consistently demonstrates encouraging performance, showing the strength and effectiveness in conducting the semi-supervised RS change detection task.

2) Strategy Comparison: We present the results of the strategy comparison in Table II. To clarify the improvement brought by the semi-supervised strategy over the backbone, we provide the results of the backbone denoted as "SupOnly," which is trained solely using the labeled data $\mathcal{D}_l$. According to the results, our strategy ECPS obviously achieves the best performance on four datasets. Compared with the original version of CPS, ECPS occupies the significant advantages when facing limited labeled data, such as 5% and 10%. As the labeled data increase from 5% to 40%, ECPS always outperforms the CPS, although the performance difference is reduced. In addition, it is worth noting that the addition of

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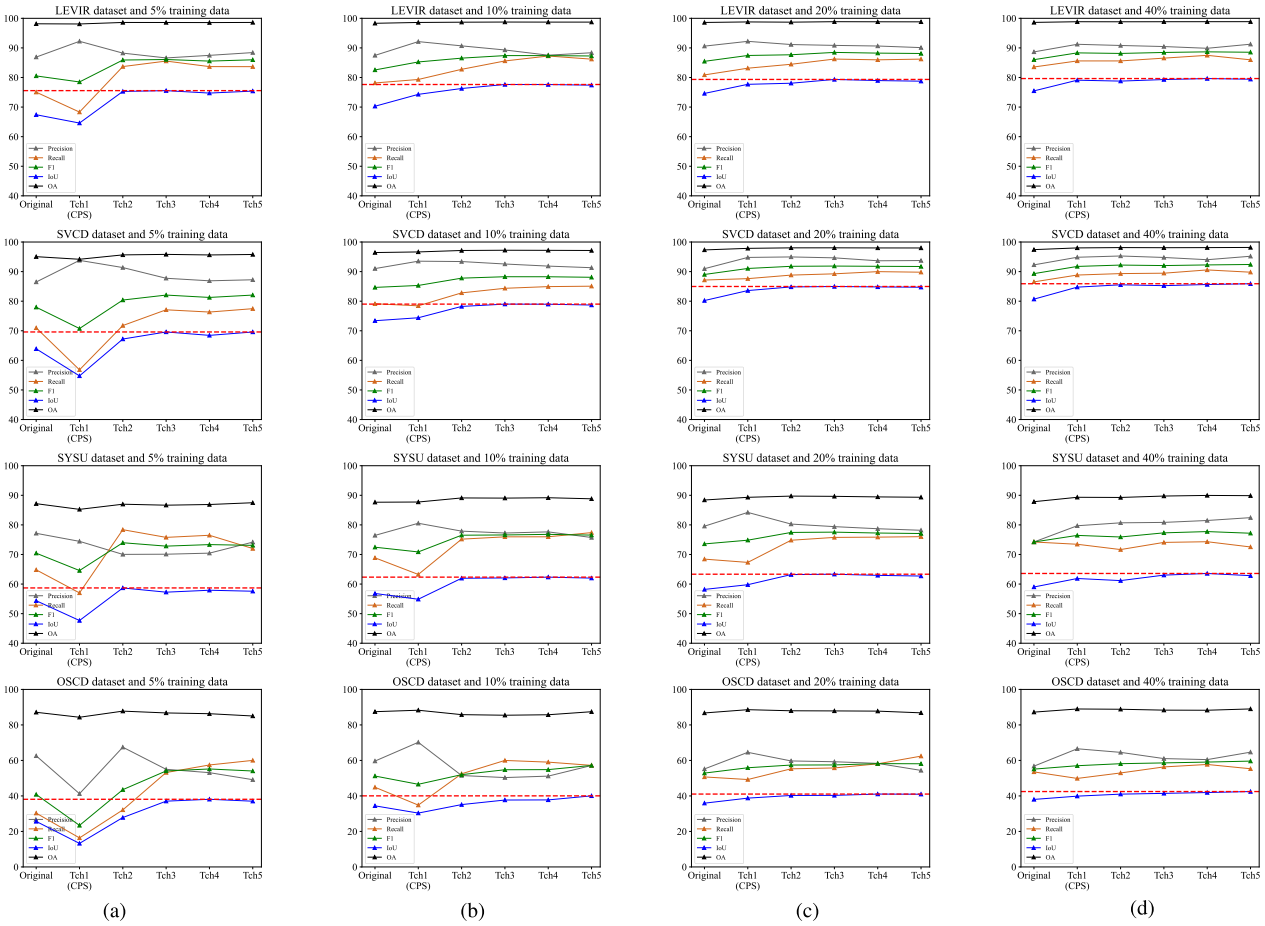


Fig. 4. Parameter analysis of different teacher models. The first, second, third, and fourth rows correspond to the results of LEVIR, SVCD, SYSU, and OSCD datasets, respectively. Here, "original" denotes the results obtained from the backbone trained solely with labeled data $\mathcal{D}_l$, and "Tch$x$" means the teacher model trained with x student models in the ensemble. In addition, we use a red line in each figure to indicate the model with the highest IoU value, making it easier to identify the top-performing model. (a) 5%. (b) 10%. (c) 20%. (d) 40%.

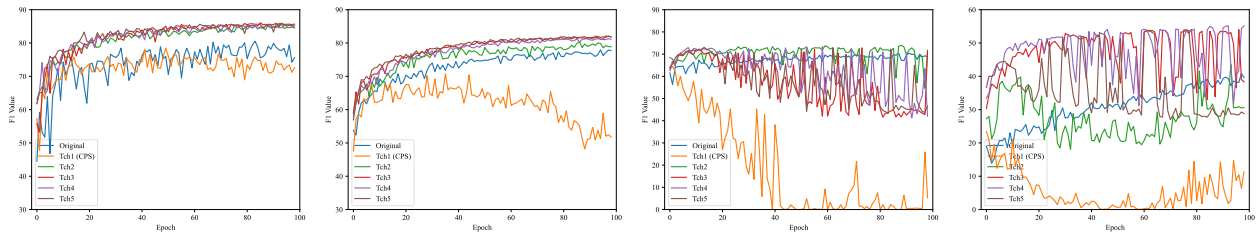


Fig. 5. Accuracy change of different teacher models at the 5% training ratio of four datasets. Here, "Tch i" represents the teacher model trained with i student models, and "original" denotes the backbone model trained solely with labeled data $\mathcal{D}_l$. (a) LEVIR. (b) SVCD. (c) SYSU. (d) OSCD.

certain semi-supervised strategies leads to a decrease in the backbone performance, such as CCC and CPS at 5%. This is because the limited labeled data result in the poor backbone performance, which significantly interferes with the original capacity of the backbone model obtained from the labeled data, ultimately causing a decline in its performance. In brief, the strategy comparison results provide strong evidence about the superiority of ECPS for conducting semi-supervised task.

C. Parameter Analysis

In our model, two hyperparameters, δ for adjusting model's tendency toward precision or recall and the number of student models n, require manual configuration. Therefore, we investigate their influence on model performance from the following three aspects: 1) the impact of n on teacher model's performance; 2) the influence of δ on the precision and recall of prediction results generated by the teacher model; and 3) the effect of n on the single student model's performance.

1) $n \rightarrow$ Teacher Model: To clearly show the influence of student number n on teacher model's performance, we select five values ranging from 1 to 5 (the maximum our server can handle). It is important to note that when n is set to 1, ECPS will degenerate into CPS, as there will not be several student models for ensemble. We present all experimental results of four datasets with four different training ratios in Fig. 4.

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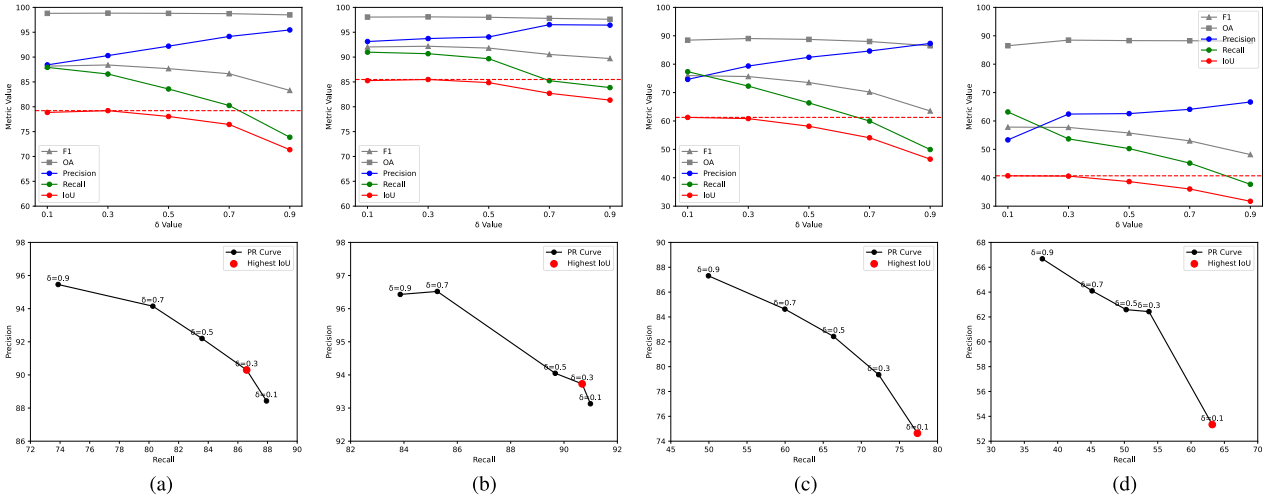


Fig. 6. Variation of metrics based on different hyperparameter values of δ, allowing us to adjust the model's tendency toward precision or recall. (a) LEVIR. (b) SVCD. (c) SYSU. (d) OSCD.

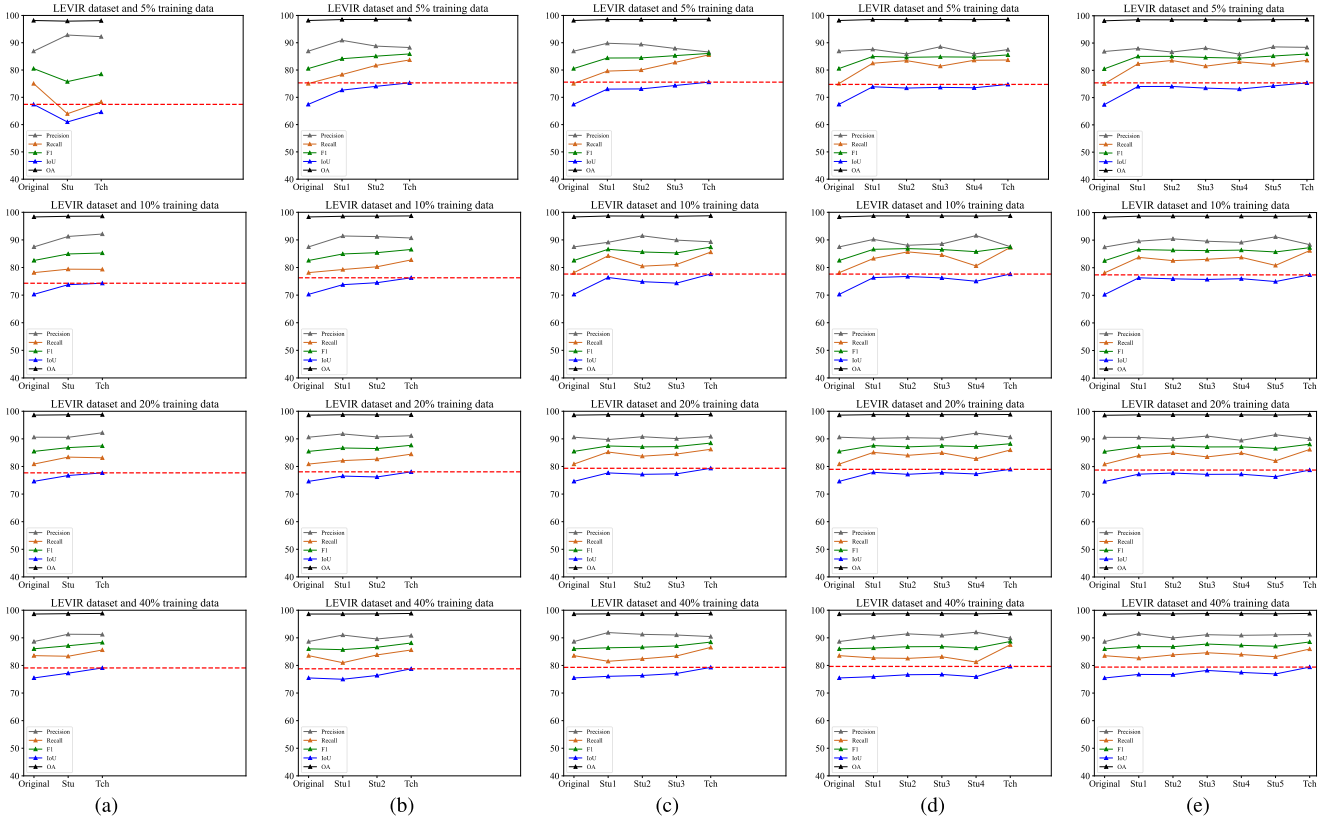
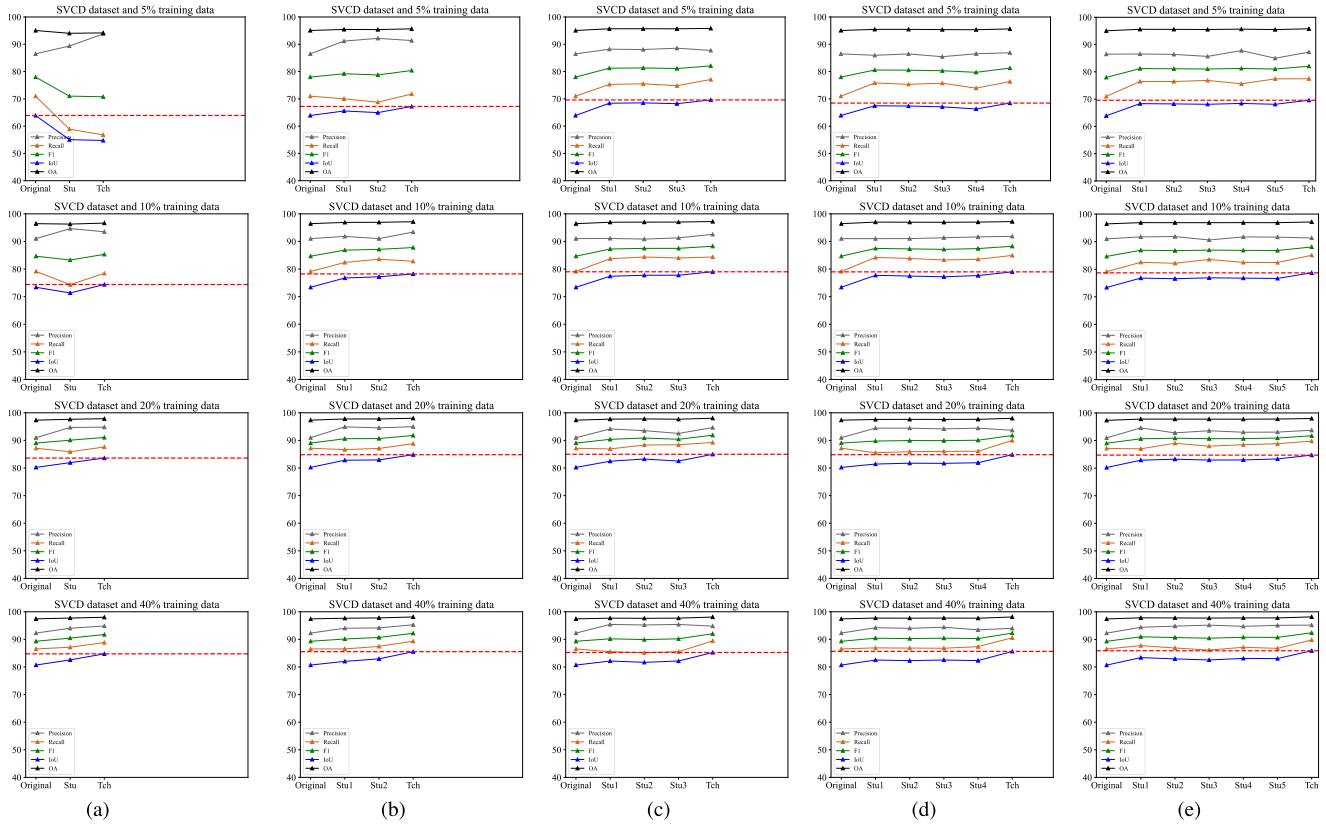


Fig. 7. Parameter analysis of the LEVIR dataset based on different student numbers and training ratios. Here, "Stu i" denotes the ith student model, and "Tch" represents the corresponding teacher model. The red line marks the highest IoU value for indicating the top-performing model. (a) One student (CPS). (b) Two students. (c) Three students. (d) Four students. (e) Five students.

For clarity, we label the teacher model as "Tch$x$," where x represents the number of student models used during training. For example, "Tch1" denotes the teacher model trained with only one student model, representing CPS. The "original" results obtained from the backbone trained solely with labeled data $\mathcal{D}_l$, without any student model (i.e., zero student models).

By observing the results, two notable points become apparent. First, as the student number increases, from "original" to "Tch3," the performance of teacher models has a rising trend. However, as we move from "Tch3" to "Tch5," the performance tends to stabilize, indicating a reduced impact on further improvement with additional student models. Therefore, setting n to 3 can be a reasonable configuration for balancing computational cost and performance improvement. Second, when dealing with the limited labeled data, such as 5% and 10%, the "Tch1 (CPS)" model exhibits worse performance than "original" model, which is caused by the many noises within pseudo-labels. To clarify this process,

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(a) (b) (c) (d) (e)

Fig. 8. Parameter analysis of the SVCD dataset based on different student numbers and training ratios. Here, "Stu i" denotes the ith student model, and "Tch" represents the corresponding teacher model. The red line marks the highest IoU value for indicating the top-performing model. (a) One student (CPS). (b) Two students. (c) Three students. (d) Four students. (e) Five students.

we plot the $F1$ change lines for "original" and from "Tch1" to "Tch5" in Fig. 5, under the training ratio of 5% on four datasets. As the training epochs increase, we observe that from "Tch2" to "Tch5," there is a consistent upward trend for $F1$ scores on LEVIR, SVCD, and OSCD datasets. However, upon completing the full training, the $F1$ of "Tch1 (CPS)" shows a significant decrease on SVCD and SYSU datasets, showing the negative influence from noises within pseudo-labels. This finding means that when facing limited labeled data, n should be set to more than 1 for ensuring performance improvement.

2) $\delta \rightarrow$ Precision and Recall: In light of practical application requirements, we introduce a hyperparameter δ in prediction ensemble strategy to control the model's tendency toward precision or recall. Since δ cannot equal 0 or 1 (see Section III-C), we choose five values from 0.1 to 0.9 with the interval of 0.2 and plot the corresponding results along with the precision and recall curve for four datasets in Fig. 6, as illustrated. Here, the results are generated based on the training ratio of 20%. According to the results, the larger/smaller δ value leads to the higher/lower precision and the lower/higher recall. This clearly demonstrates the effectiveness of δ in adjusting the model's tendency. Furthermore, we can find that the model achieves the best overall performance when δ equals 0.3. Therefore, different δ values can be chosen to satisfy the various practical requirements, such as 0.1/0.3/0.9 for higher recall/overall performance/precision.

Last but not least, let us explain why a high δ value leads to low overall performance, while a low δ value does not. This can be mainly attributed to the category imbalance (unchanges much more than changes), which can be observed in most bitemporal image pairs. Due to the category imbalance, the model's predictions tend to contain more unchanged pixels with the value of 0. As a result, high δ value further pushes abundant pixels of $\mathbf{Y}_e$ to 0 (unchange), thereby producing many fake unchanges and significantly decreasing model's overall performance. On the contrary, a low δ value, with its prediction bias toward 0, has difficulty pushing a large number of pixels to the value of 1 (change). Therefore, the model generates fewer fake changes, resulting in little overall influence on the model's performance.

3) $n \rightarrow$ Student Models: We show the 4×5 experimental results of LEVIR/SVCD/SYSU/OSCD datasets in Figs. 7–10, respectively, which consists of four different training ratios and five different student numbers. In each case, we name the ith student model as "Stu i," and the corresponding teacher model as "Tch." According to the results, the performance of teacher model is always superior to that of student models, since the teacher model is trained based on ensemble pseudo-labels that contains fewer noises. As the student number n increases from 1 to 3, the performances of student models are also slightly improved, as they are supervised with the pseudo-label $\mathbf{Y}$ generated by the stronger teacher model. During moving from three students to five students, the performances of student

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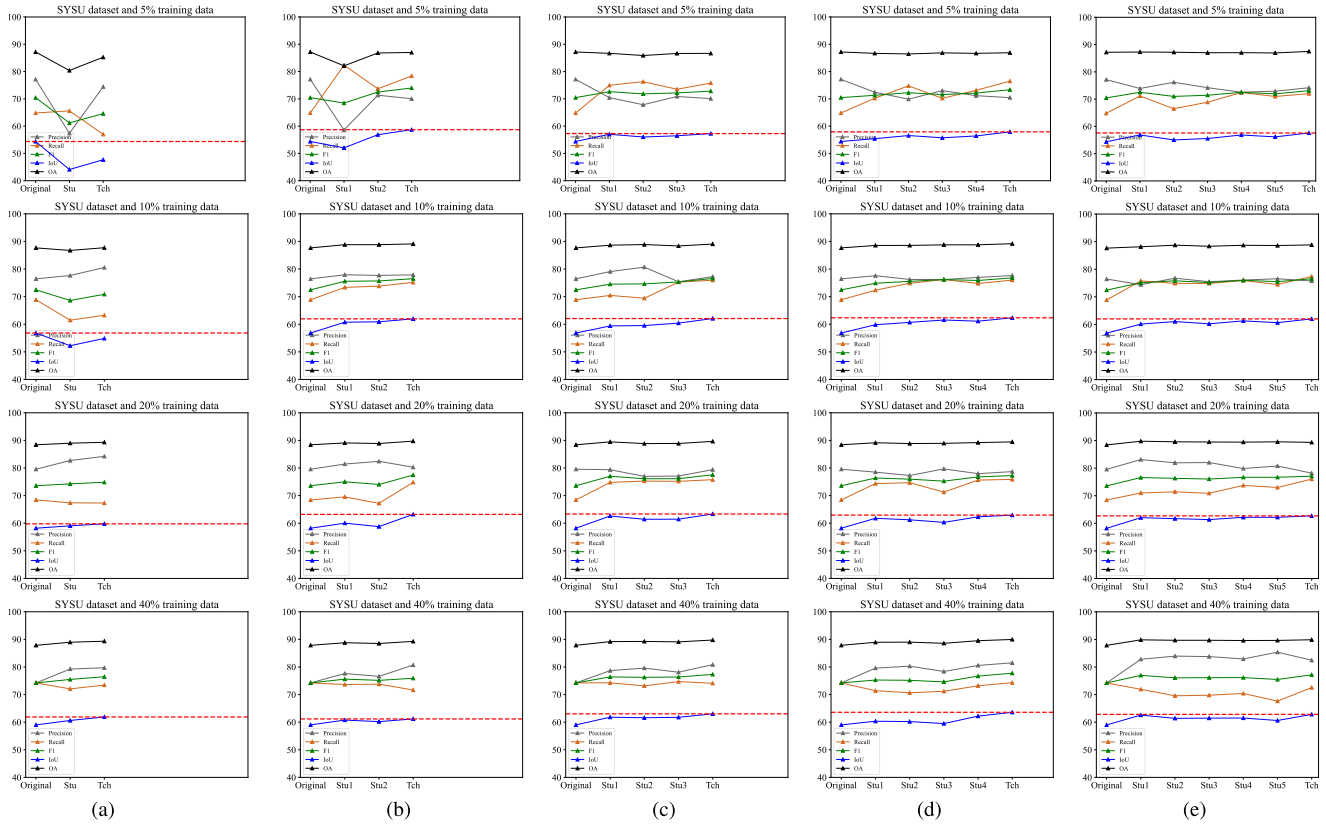


Fig. 9. Parameter analysis of the SYSU dataset based on different student numbers and training ratios. Here, "Stu i" denotes the ith student model, and "Tch" represents the corresponding teacher model. The red line marks the highest IoU value for indicating the top-performing model. (a) One student (CPS). (b) Two students. (c) Three students. (d) Four students. (e) Five students.

models tend to stability. In summary, the influence of n on the student models is similar to that of teacher model, since the crosswise supervision manner of ECPS leads to a crosswise performance effect between student and teacher models.

D. Ablation Studies

In all experiments, we calculate the supervised loss in the labeled data $\mathcal{D}_l$, referred to as $\mathcal{L}^l_{\text{sup}}$. Meanwhile, based on ECPS, the unsupervised loss is computed in the unlabeled data $\mathcal{D}_u$, denoted as $\mathcal{L}^u_{\text{ecps}}$. This independent loss calculation ensures the integrity of the experiment and clearly demonstrates the effectiveness of ECPS in utilizing the unlabeled data $\mathcal{D}_u$. Furthermore, according to the experiments conducted with CPS in [35], applying unsupervised CPS to the labeled data $\mathcal{D}_l$ can further improve the model performance. Therefore, we introduce similar experiments, i.e., applying ECPS to $\mathcal{D}_l$, and mark the corresponding loss as $\mathcal{L}^l_{\text{ecps}}$. With these three losses, $\mathcal{L}^l_{\text{sup}}$, $\mathcal{L}^l_{\text{ecps}}$, and $\mathcal{L}^u_{\text{ecps}}$, we perform the ablation studies and list the results in Table III. It is easy to find that adding either $\mathcal{L}^l_{\text{ecps}}$ or $\mathcal{L}^u_{\text{ecps}}$ to $\mathcal{L}^l_{\text{sup}}$ (i.e., $\mathcal{L}^l_{\text{sup}} \leftarrow \mathcal{L}^l_{\text{ecps}}$ or $\mathcal{L}^l_{\text{sup}} \leftarrow \mathcal{L}^u_{\text{ecps}}$) significantly improves the model performance on all four datasets. Specially, $\mathcal{L}^l_{\text{sup}} \leftarrow \mathcal{L}^u_{\text{ecps}}$ shows remarkable enhancement, while $\mathcal{L}^l_{\text{sup}} \leftarrow \mathcal{L}^l_{\text{ecps}}$ leads to slight improvement. Moreover, when adding $\mathcal{L}^l_{\text{ecps}}$ to our method (i.e., $\mathcal{L}^l_{\text{sup}} + \mathcal{L}^u_{\text{ecps}} \leftarrow \mathcal{L}^l_{\text{ecps}}$), the model performance is slightly enhanced on two datasets, LEVIR and SYSU. This find proves that applying ECPS to both $\mathcal{D}_l$ and $\mathcal{D}_u$ can further improve model performance, with the effectiveness of applying ECPS to $\mathcal{D}_u$ being more prominent than that of $\mathcal{D}_l$.

E. Cost Analysis of Time, Storage, and Computation

In this section, we conduct an analysis of the model cost during the training and inference stage, considering factors, such as time for training and time, storage, and computation for inference. Here, the SVCD dataset with a training ratio of 5% is selected to show the training cost and test accuracy (IoU). Based on comprehensive considerations, four methods (RCL [38], s4GAN [39], AdvNet [41], and Revisiting [37]) from Section IV-B1 and four strategies (PS [48], MT [47], CCC [46], and CPS [35]) from Section IV-B2 are chosen for cost evaluation. It is worth noting that applying different semi-supervised strategies (e.g., PS, MT, CCC, CPS, and our ECPS with different student numbers) does not alter the model's inference cost. The comparison results of costs are listed in Table IV, where "time" of inference denotes the processing time (in ms) for a bitemporal image pair, "time" of training indicates the time (in min) for training model, "storage" represents the file size (in MB) for storing model parameters, and "computation" means the computation complexity [in Giga Floating-point Operations Per Second (GFLOPs)] of the model. By observing the table, it is evident that the training time for "+PS" is minimal, as it employs only one model in a semi-supervised manner. In our method, the training time slightly increases with an augmented number of

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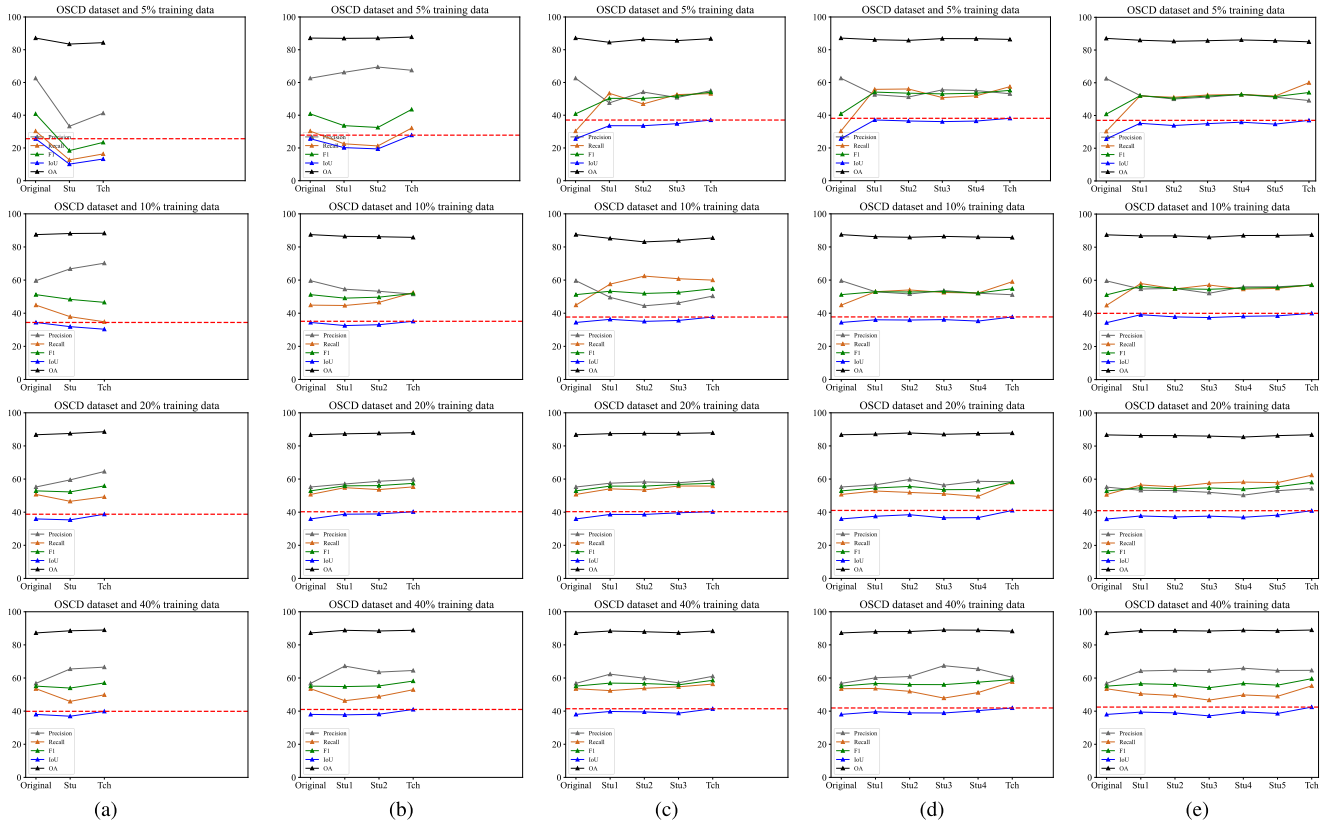


Fig. 10. Parameter analysis of the OSCD dataset based on different student numbers and training ratios. Here, "Stu i" denotes the ith student model, and "Tch" represents the corresponding teacher model. The red line marks the highest IoU value for indicating the top-performing model. (a) One student (CPS). (b) Two students. (c) Three students. (d) Four students. (e) Five students.

TABLE III
ABLATION STUDIES ON SELECTING DIFFERENT LOSSES

Losses			LEVIR					SVCD					SYSU					OSCD				
$\mathcal{L}_{sup}^{l}$	$\mathcal{L}_{ecps}^{l}$	$\mathcal{L}_{ecps}^{u}$	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
✓			90.60	80.87	85.46	74.61	98.60	90.98	87.15	89.03	80.22	97.33	79.56	68.43	73.57	58.19	88.41	55.20	50.70	52.86	35.92	86.74
✓	✓		90.81	83.51	87.01	77.00	98.73	92.54	89.27	90.88	83.28	97.77	82.78	68.26	74.82	59.77	89.17	60.05	53.24	56.44	39.31	87.95
✓		✓	90.85	84.47	87.68	78.06	98.70	94.65	89.25	**91.87**	**84.97**	**98.04**	79.41	75.77	77.55	63.33	89.65	59.26	55.76	57.46	40.31	88.59
✓	✓	✓	89.75	86.73	**88.21**	**78.91**	**98.82**	93.54	90.15	91.82	84.87	98.00	79.57	76.25	**77.87**	**63.76**	**89.78**	58.85	59.70	**59.27**	**42.12**	**88.63**

students, owing to the ensemble strategy. Nevertheless, when compared with various semi-supervised methods, our method stands out with a leading training time. Moreover, our method achieves the minimum inference cost and superior accuracy (i.e., IoU). These advantages make it a promising choice for real-world applications in semi-supervised RS change detection tasks.

F. Analysis of Pseudo-Labels

Considering the crucial role of pseudo-labels in ECPS, we further analyze the quality of pseudo-labels on the OSCD dataset with a training ratio of 20%. In addition, we provide the pseudo-label results for MT [47], CCC [46], and CPS [35] for comparison. For MT, we adopt the student model to generate the pseudo-labels. In the case of CCC and CPS, one model is selected for comparison studies, and another model is chosen to generate pseudo-labels. It is worth noting that PS [48] uses an augmented image pair to generate pseudo-labels. However, employing this strategy in the test set leads to a collapsed performance, so PS is not employed in this section. The results are listed in Table V. It is easy to see that the results for MT, CCC, and CPS are close, since the performance of the two models is similar after full training. For ECPS, the pseudo-labels are slightly better than the original result. This is because the pseudo-labels are the ensemble results of several student models, where their ensemble performance slightly exceeds that of a single teacher model. However, we suggest adopting the teacher model rather than student models in real applications, since the performance gains from multiplying inference costs are too limited.

G. Prediction Visualization

To visually evaluate the effectiveness of the proposed method, we randomly select three bitemporal image pairs

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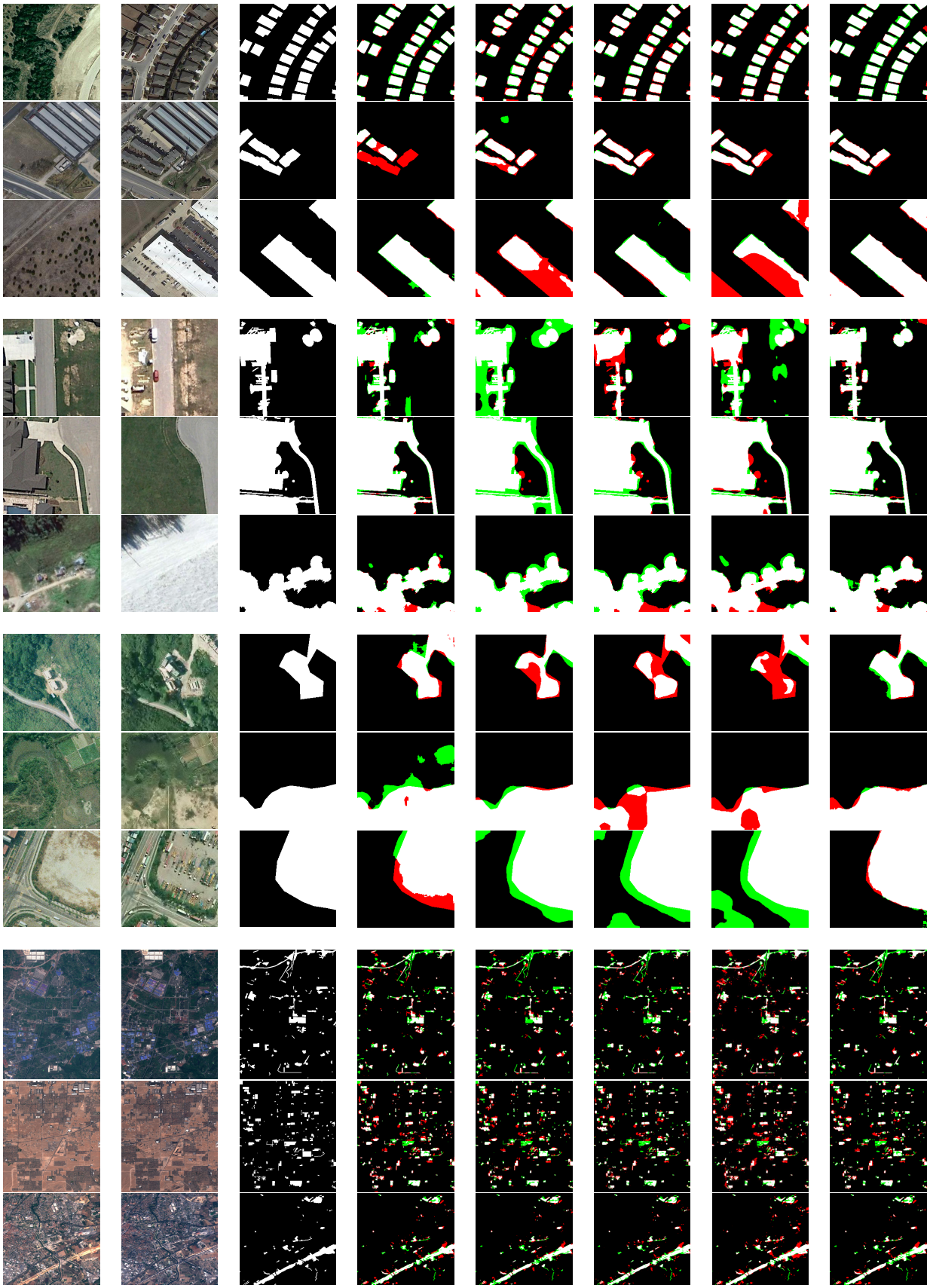


Fig. 11. Visual results of different methods on four datasets at the training ratio of 10%. Here, the TP, TN, FP, and FN pixels are marked with white, black, red, and green, respectively. Here, each group consists of three rows, corresponding to the LEVIR, SVCD, SYSU, and OSCD datasets, respectively. (a) Image T_1. (b) Image T_2. (c) Label. (d) AdvNet. (e) RCL. (f) Revisiting. (g) s4GAN. (h) Ours.

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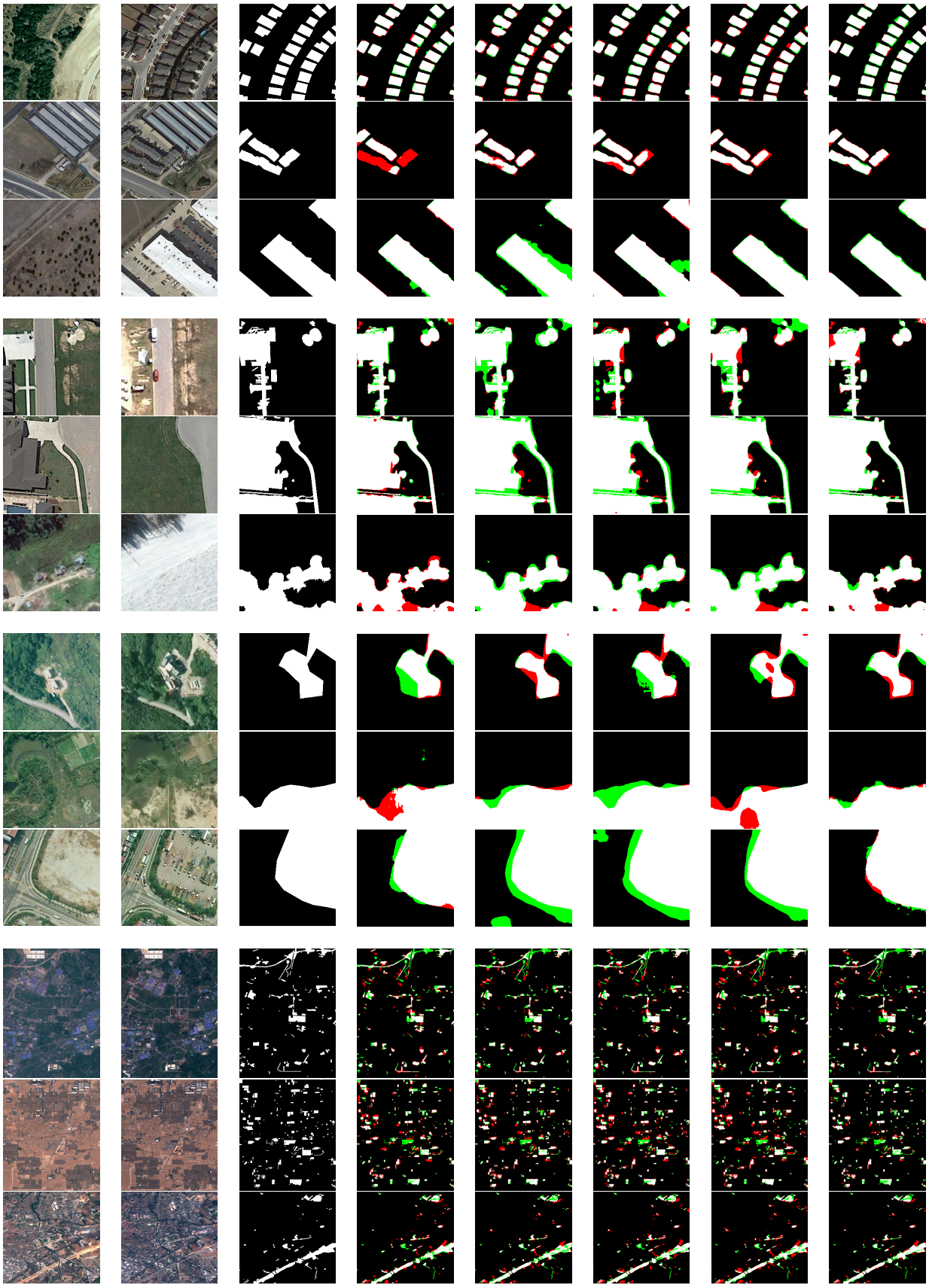


Fig. 12. Visual results of different methods on four datasets at the training ratio of 40%. Here, the TP, TN, FP, and FN pixels are marked with white, black, red, and green, respectively. Here, each group consists of three rows, corresponding to the LEVIR, SVCD, SYSU, and OSCD datasets, respectively. (a) Image T_1. (b) Image T_2. (c) Label. (d) AdvNet. (e) RCL. (f) Revisiting. (g) s4GAN. (h) Ours.

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TABLE IV
TIME, STORAGE, AND COMPUTATION OF DIFFERENT METHODS AND STRATEGIES

		Training	Inference			IoU
		Time (min) ↓	Time (ms) ↓	Storage (MB) ↓	Computation (GFLOPs)↓	
Methods	RCL [38]	453.9	35.74	188	73.23	66.40
	s4GAN [39]	338.9	32.63	171	47.47	59.98
	AdvNet [41]	331.9	40.87	175	95.75	65.13
	Revisting [37]	492.8	34.14	194	73.23	63.83
Strategies	+ PS [48]	**134.6**	4.47	15	10.85	65.53
	+ MT [47]	170.3				65.40
	+ CCC [46]	176.5				50.06
	+ CPS [35]	195.4				54.75
Ours	2 Students	260.1	**4.47**	**15**	**10.85**	67.21
	3 Students	332.6				**69.60**
	4 Students	413.2				68.45
	5 Students	498.9				69.59

TABLE V
ANALYSIS OF DIFFERENT STRATEGIES BASED ON PSEUDO-LABELS

	Pseudo Labels	Pre	Rec	**F1**	**IoU**	**OA**
MT [47]	×	59.96	45.08	51.47	34.65	87.53
	✓	59.73	44.80	51.20	34.41	87.48
CCC [46]	×	60.28	47.63	53.22	36.25	87.72
	✓	60.55	47.38	53.16	36.20	87.76
CPS [35]	×	64.56	49.24	55.87	38.76	87.89
	✓	64.99	48.82	55.75	38.65	87.72
Ours	×	59.26	55.76	57.46	40.31	88.59
	✓	58.33	58.08	58.20	41.05	88.63

from each dataset for exhibition. The prediction results of our method and four comparative methods at 10% training ratios are shown in Fig. 11. In addition, we also show the results with the training ratio of 40% in Fig. 12. To enhance clarity, we use white, black, red, and green colors to represent true positive (TP), true negative (TN), false positive (FP), and false negative (FN) pixels, respectively. According to the figure, our results closely align with the ground truth labels and exhibit the fewest pixels with false predictions (i.e., FP and FN), indicating highly accurate visual predictions. Moreover, despite under the supervision of a limited labeled data (only 10%), our method effectively handles the edge regions of land covers and produces more detailed change information. These visual findings provide robust evidence of the proposed method's superiority in performing the semi-supervised RS change detection task.

V. CONCLUSION AND FUTURE WORKS

In this article, we successfully apply ensemble learning to the semi-supervised strategy CPS, creating ECPS. Based on ECPS, a method is proposed to conduct the semi-supervised RS change detection task. In this method, a teacher model and n ensemble student models are created, with applying the crosswise pseudo supervision between teacher and student models. Here, the labeled data adopts the conventional supervised training with labels, while ECPS is applied to the unlabeled data for an unsupervised training. The extensive experiments are conducted on four datasets with four different training ratios, strongly demonstrating its superiority and effectiveness in performing the semi-supervised RS change detection task.

The success of ECPS primarily lies in its ability to improve the quality of pseudo-labels. Therefore, our future work will focus on employing ensemble learning to optimize other semi-supervised strategies that rely on pseudo-labels, while simultaneously reducing the cost caused by ensemble learning. Furthermore, we plan to explore the application of ECPS to other semi-supervised tasks and assess its universality across different domains.

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